ELECTRONIC PRODUCT DESIGN & REVERSE ENGINEERING

CMOS NOR Gate

Construction, Simulation and Truth Table using MOS Transistors in Tinkercad

Project	Project Part B
Question	5b - Construct and simulate a NOR gate using MOS transistors, infer its behavior, and explain its truth table with logical operation
Group Number	**31**
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1. Aim

To build a 2-input NOR gate using two PMOS and two NMOS transistors in Tinkercad Circuits, simulate it for all four input combinations, and verify the truth table Y = (A + B)'.

2. Components Required

Ref	Component (Tinkercad name)	Qty	Value / setting
MP1, MP2	P-channel MOSFET	2	Search 'MOSFET' in the components panel
MN1, MN2	N-channel MOSFET	2	Search 'MOSFET' in the components panel
PS1	Power Supply	1	Set to 5 V
SW_A, SW_B	Slide Switch (3-pin)	2	Input switches A and B
R1	Resistor	1	330 ohm (LED current limit)
LED1	LED (any colour)	1	Output indicator: ON = logic 1
M1 (optional)	Multimeter	1	Set to DC voltage; probe Y to GND
BB1	Breadboard	1	Red rail = VDD, blue rail = GND

Tip: hover over each MOSFET pin in Tinkercad to confirm which leg is Drain (D), Gate (G) and Source (S) before wiring.

3. Schematic to Follow

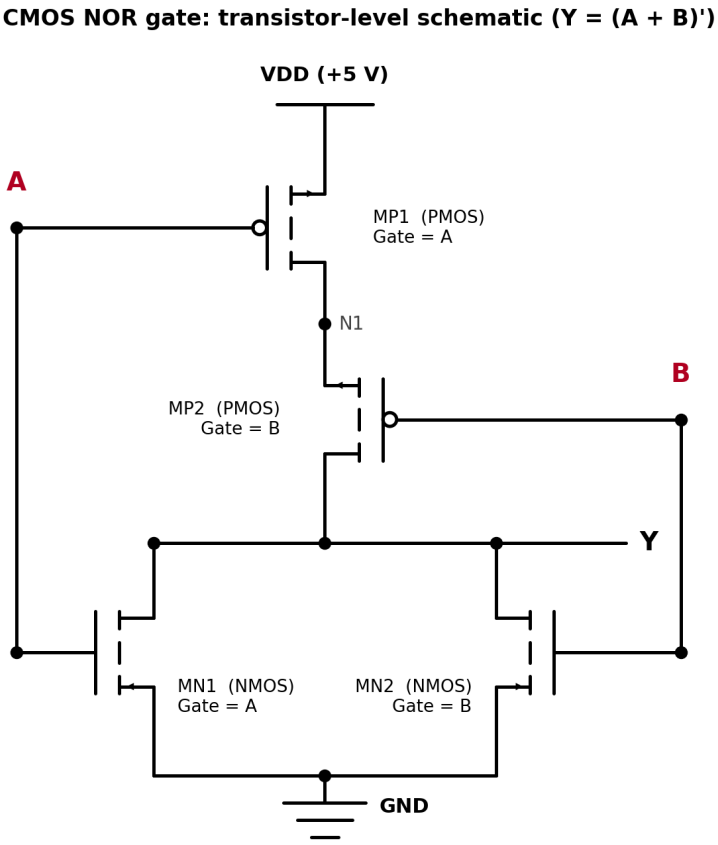


Figure 1: Transistor-level schematic of the 2-input CMOS NOR gate (use the node names A, B, N1, Y in the tables).

Device	Drain (D)	Gate (G)	Source (S)	Role
MP1 (PMOS)	N1	A	VDD	Series pull-up
MP2 (PMOS)	Y	B	N1	Series pull-up
MN1 (NMOS)	Y	A	GND	Parallel pull-down
MN2 (NMOS)	Y	B	GND	Parallel pull-down

4. Connections (copy-ready)

4.1 Net list. Every pin listed on one line must be joined together.

Net	Pins connected to this net	Wire colour
VDD	PS1 (+), MP1 Source, SW_A pin 1, SW_B pin 1	Red
GND	PS1 (-), MN1 Source, MN2 Source, SW_A pin 3, SW_B pin 3, LED1 cathode	Black
A	SW_A middle pin, MP1 Gate, MN1 Gate	Blue
B	SW_B middle pin, MP2 Gate, MN2 Gate	Green
N1	MP1 Drain, MP2 Source (internal node, nothing else)	Purple
Y	MP2 Drain, MN1 Drain, MN2 Drain, R1 pin 1	Orange
LED	R1 pin 2, LED1 anode	Orange

4.2 Wire-by-wire list. Tick each wire as you place it.

No.	From	To	Done
1	PS1 (+)	VDD rail	[ ]
2	PS1 (-)	GND rail	[ ]
3	VDD rail	MP1 Source	[ ]
4	MP1 Drain	MP2 Source	[ ]
5	MP2 Drain	Y	[ ]
6	MN1 Drain	Y	[ ]
7	MN2 Drain	Y	[ ]
8	MN1 Source	GND rail	[ ]
9	MN2 Source	GND rail	[ ]
10	SW_A pin 1	VDD rail	[ ]
11	SW_A pin 3	GND rail	[ ]
12	SW_A middle pin	A	[ ]
13	A	MP1 Gate	[ ]
14	A	MN1 Gate	[ ]
15	SW_B pin 1	VDD rail	[ ]
16	SW_B pin 3	GND rail	[ ]
17	SW_B middle pin	B	[ ]
18	B	MP2 Gate	[ ]
19	B	MN2 Gate	[ ]
20	Y	R1 pin 1	[ ]
21	R1 pin 2	LED1 anode	[ ]
22	LED1 cathode	GND rail	[ ]

5. Build Steps in Tinkercad

1. Open tinkercad.com, go to **Circuits**, and click **Create new Circuit**. Rename it 'CMOS_NOR_Q5b'.

2. From the components panel, drag in the breadboard, power supply, 2 slide switches, resistor, LED, and 2 P-channel plus 2 N-channel MOSFETs.

3. Click the power supply and set its voltage to **5 V**. Click the resistor and set it to **330 ohm**.

4. Connect PS1 (+) to the red rail and PS1 (-) to the blue rail (wires 1 and 2).

5. Place the four MOSFETs on the breadboard with space between them. Label them MP1, MP2, MN1, MN2 using the Notes tool.

6. Make the pull-up network first (wires 3 to 5), then the pull-down network (wires 6 to 9).

7. Wire the two switches (wires 10 to 12 and 15 to 17), then connect A and B to the gates (wires 13, 14, 18, 19).

8. Wire the output indicator (wires 20 to 22). Optionally attach the multimeter between Y and the GND rail.

9. Click **Start Simulation**. Click each switch to set A and B to the four combinations and note the LED and Y voltage in the table below.

10. Take screenshots of the full circuit and of each simulation state for your report.

6. Expected Results and Observation Table

Switch position: pin 1 side = logic 1 (VDD), pin 3 side = logic 0 (GND). Fill the last two columns from your simulation.

A	B	Expected Y	Expected LED	Observed Y (V)	Observed LED
0	0	1 (about 5 V)	ON		
0	1	0 (about 0 V)	OFF		
1	0	0 (about 0 V)	OFF		
1	1	0 (about 0 V)	OFF		

7. Inference

Y is high only when A = B = 0, because only then both series PMOS devices conduct and both parallel NMOS devices are off. If either input is 1, at least one NMOS conducts and pulls Y to GND, and the series PMOS path is broken. This is the NOR function, Y = (A + B)' = A'.B'.

8. Simulation Results

The figures below were generated from a switch-level simulation of the circuit in Figure 1: each MOSFET is ON or OFF according to its gate input, and Y is pulled to VDD or GND by whichever network conducts.

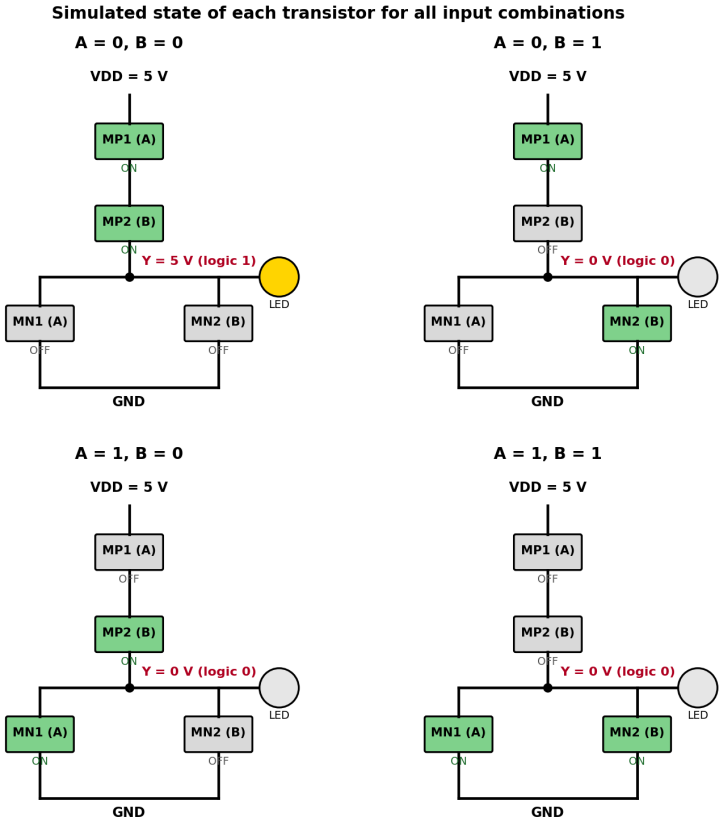


Figure 2: Transistor states, LED and output Y for A, B = 00, 01, 10, 11.

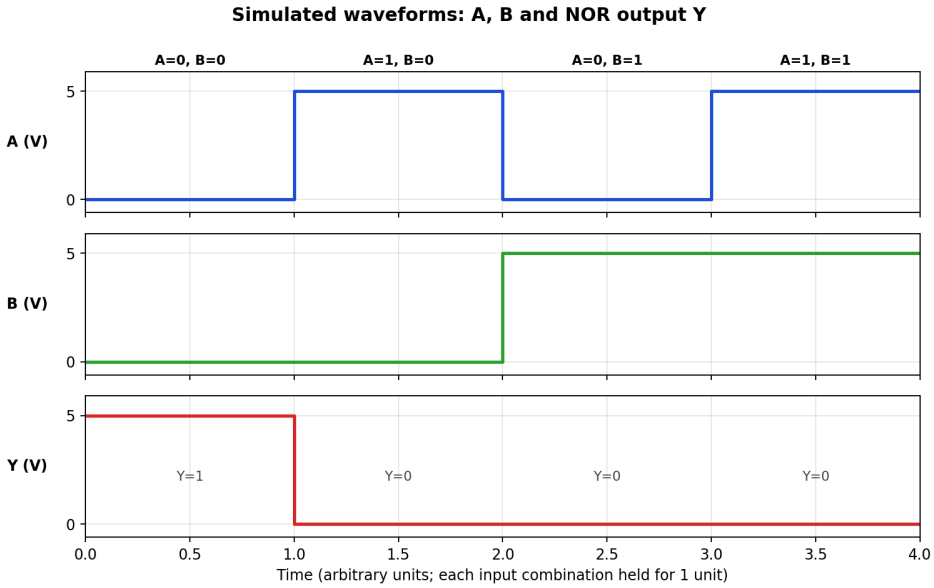


Figure 3: Waveforms of A, B and Y. Y is high only while A = B = 0.

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